

Computational Mechanics:

3. Prediction

ILIAD INTENSIVE - SUMMER 2026

Outline :

- 1) Belief state
- 2) Next-token (probability) vector
- 3) Mixed-state presentation
- 4) GHMM prediction

Belief state

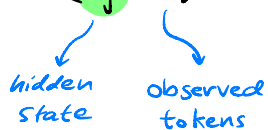
How should we predict tokens emitted from an HMM?

↳ By defⁿ the next-token distribution is a function of the current distribution over hidden states.

↳ Use this to make predictions

Bayes' rule tells us that the posterior prob. of occupying state s_j after observing x is:

$$P(s_j | x) = \frac{P(x, s_j)}{P(x)} = \frac{\sum_{s_0 \in S} P(s_0) P(x, s_j | s_0)}{P(x)}$$



hidden state observed tokens

Similarly, posterior prob. of occupying s_j after observing the sequence $W = w_1, w_2 \dots w_n$ is:

$$P(s_j | W) = \frac{\sum_{s^0, \dots, s^{n-1}} P(s^0) P(w_1, s^1 | s^0) P(w_2, s^2 | s^1) \dots P(w_n, s_j | s^{n-1})}{P(W)}$$

Expressing this object in terms of HMM data:

$$P(s_j | W) = \left(\frac{\eta(\phi) T(w_1) T(w_2) \dots T(w_n)}{\underbrace{\eta(\phi) T(w_1) T(w_2) \dots T(w_n)}_{\text{row-vector}} \gamma} \right)_j$$

where we order the hidden states $S = \{s_0, s_1, \dots\}$.

This motivates the defⁿ of the belief state:

$$\eta^{(w)} := \frac{\eta^{(\emptyset)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)}}{\eta^{(\emptyset)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} \gamma}, \quad w = w_1 w_2 \dots w_n \in \mathcal{X}^*$$

whose j^{th} -element corresponds to the prob. that the hidden state is s_j after obs. w .

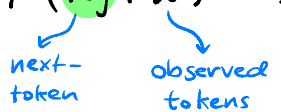
Suppose we observe a further token $x \in \mathcal{X}$, the predictive vector update rule is simple:

$$\eta^{(w)} \mapsto \eta^{(wx)} = \frac{\eta^{(w)} T^{(x)}}{\eta^{(w)} T^{(x)} \gamma}$$

Next-token (probability) vector

For predicting next-token emissions, we actually care about:

$$P(x_j | w) = \frac{P(w x_j)}{P(w)} = \frac{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} T^{(x_j)} \gamma}{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} \gamma}$$



This motivates the defⁿ of the next-token vector:

$$p^{(w)} := \left(\frac{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} T^{(x_j)} \gamma}{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} \gamma} \right)_{j=0,1,\dots}$$

where we order the tokens $\mathcal{X} = \{x_0, x_1, \dots\}$.

The j^{th} -element of $p^{(w)}$ corresponds to the prob. that the next-token is x_j after obs. w .

The belief state appears directly in the defⁿ of the next-token vector:

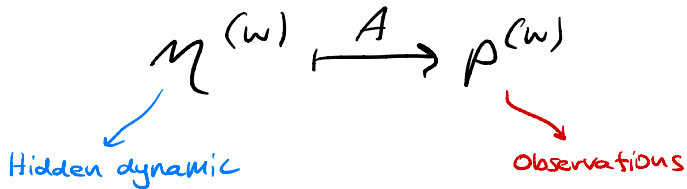
$$p_j^{(w)} := \frac{\eta^{(\emptyset)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} T^{(x_j)} \gamma}{\eta^{(\emptyset)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} \gamma} = \eta^{(w)} T^{(x_j)} \gamma$$

It follows that there exists a matrix that maps $\eta^{(w)} \mapsto p^{(w)}$ i.e.:

$$p^{(w)} = \eta^{(w)} A$$

where the matrix elements of A are:

$$A_{ij} := \sum_{k=0}^{|S|-1} T_{ik}^{(x_j)}, \quad \begin{array}{l} i = 0, 1, \dots, |S|-1 \\ j = 0, 1, \dots, |X|-1 \end{array}$$



If $\ker(A) = \emptyset$:



If $\ker(A) \neq \emptyset$:



Belief states are also useful for all future preds.

Let $w^{(p)} = w_1^{(p)} w_2^{(p)} \dots w_m^{(p)}$ & $w^{(f)} = w_1^{(f)} w_2^{(f)} \dots w_n^{(f)}$:

$$\begin{aligned} P(w^{(f)} | w^{(p)}) &= \frac{\eta^{(\emptyset)} T^{(w_1^{(p)})} \dots T^{(w_m^{(p)})} T^{(w_1^{(f)})} \dots T^{(w_n^{(f)})} \mathbf{1}}{\eta^{(\emptyset)} T^{(w_1^{(p)})} \dots T^{(w_m^{(p)})} \mathbf{1}} \\ &= \eta^{(w^{(p)})} T^{(w_1^{(f)})} \dots T^{(w_n^{(f)})} \mathbf{1} \end{aligned}$$

As in the next-token case there exists a multi-linear map from $\eta^{(w^{(p)})}$ to all future predictions.

Mixed-state presentation

For any HMM, the deterministic update rule $\eta^{(w)} \mapsto \eta^{(wx)}$ defines a meta-dynamic over beliefs called the mixed-state presentation (MSP).

The MSP is itself a (unifilar) HMM:

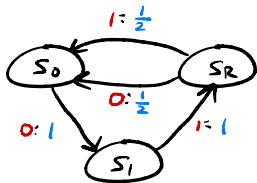


whose hidden states are the belief states.

[For unifilar HMMs, an ε -machine is the MSP of the corresponding minimal HMM.]

Example 1: (zero-one-random $p = \frac{1}{2}$)

Recall:



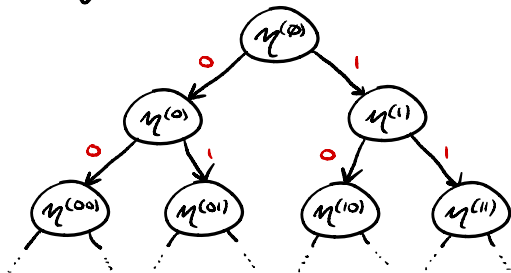
$$T^{(0)} = \begin{matrix} & \begin{matrix} S_0 & S_1 & S_R \end{matrix} \\ \begin{matrix} S_0 \\ S_1 \\ S_R \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 \end{bmatrix} \end{matrix}$$

$$T^{(1)} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & 0 & 0 \end{bmatrix}$$

$$\eta^{(w)} = \frac{\eta^{(\emptyset)} T^{(w)}}{\eta^{(\emptyset)} T^{(w)} \mathbf{1}}$$

$$T^{(w)} := T^{(w_1)} \dots T^{(w_n)}$$

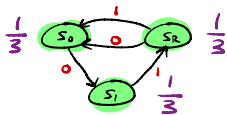
Deriving the MSP compute:



Identify nodes
where

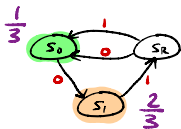
$$\eta^{(w)} = \eta^{(w')}$$

Fix: $\mu^{(\emptyset)} = [\frac{1}{3} \quad \frac{1}{3} \quad \frac{1}{3}]$

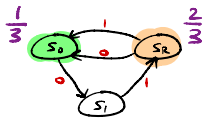


and compute:

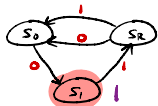
$$\mu^{(0)} = [\frac{1}{3} \quad \frac{2}{3} \quad 0]$$



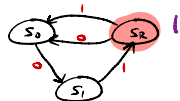
$$\mu^{(1)} = [\frac{1}{3} \quad 0 \quad \frac{2}{3}]$$



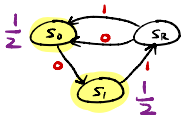
$$\mu^{(\infty)} = [0 \quad 1 \quad 0]$$



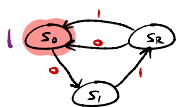
$$\mu^{(01)} = [0 \quad 0 \quad 1]$$



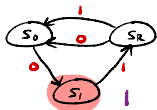
$$\mu^{(10)} = [\frac{1}{2} \quad \frac{1}{2} \quad 0]$$



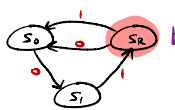
$$\mu^{(11)} = [1 \quad 0 \quad 0]$$



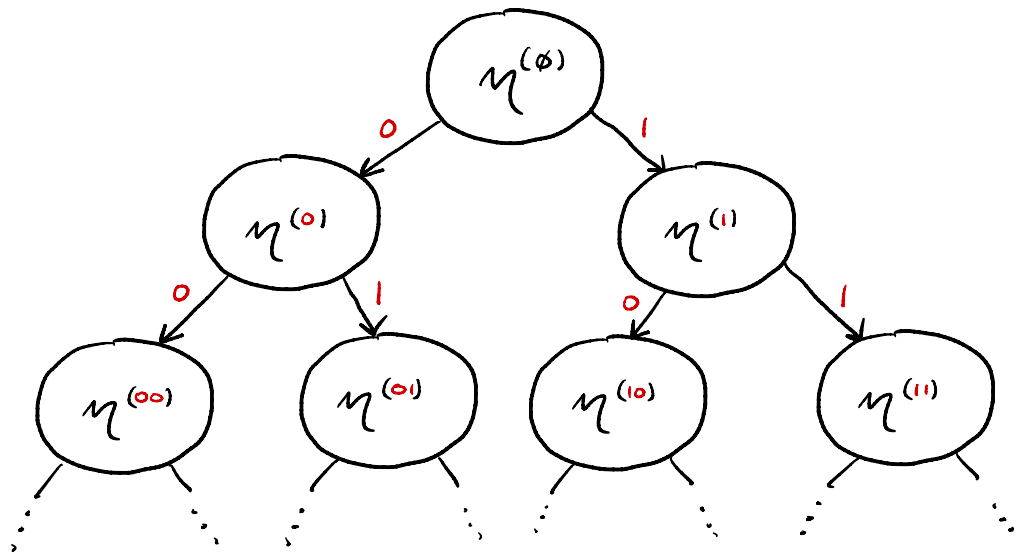
$$\mu^{(100)} = [0 \quad 1 \quad 0]$$

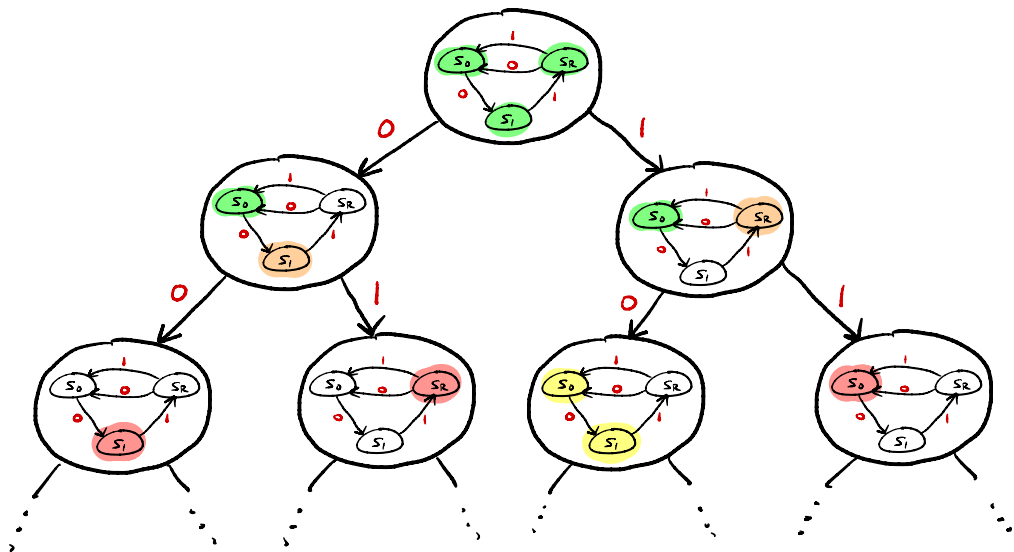


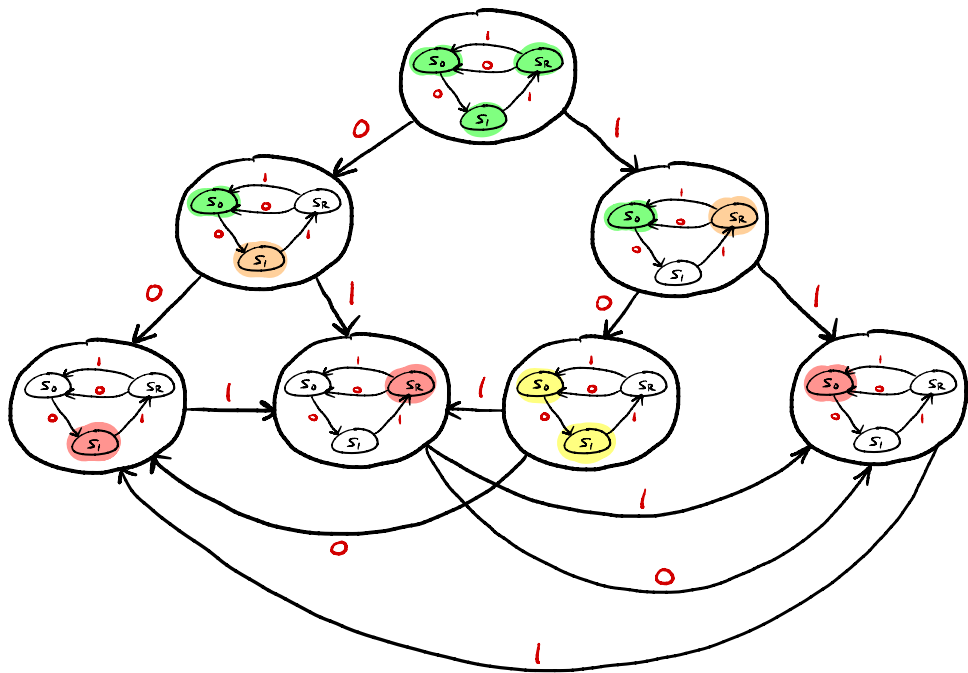
$$\mu^{(101)} = [0 \quad 0 \quad 1]$$

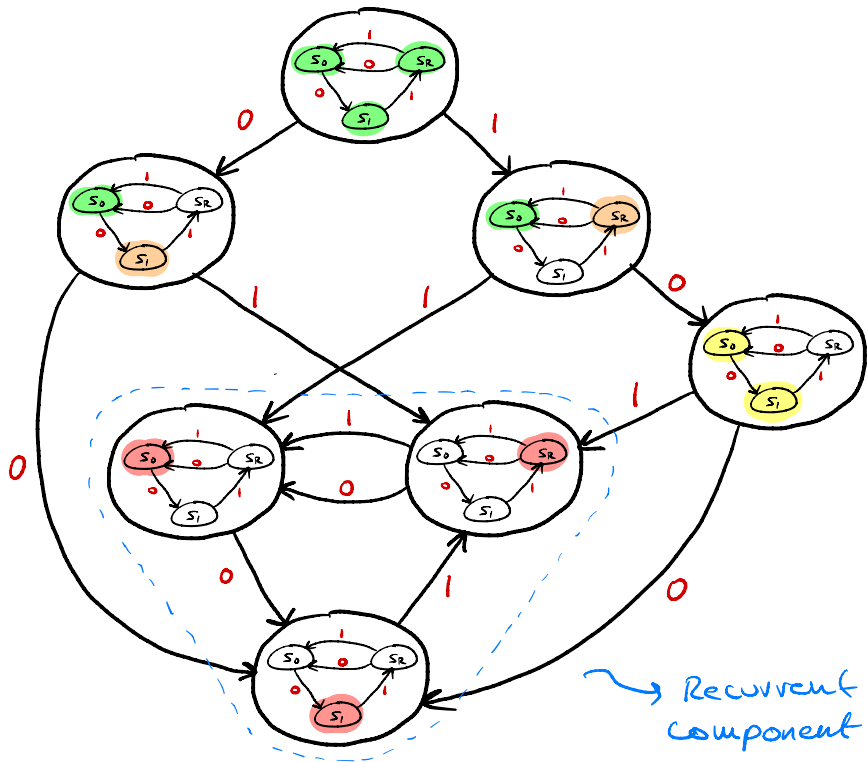


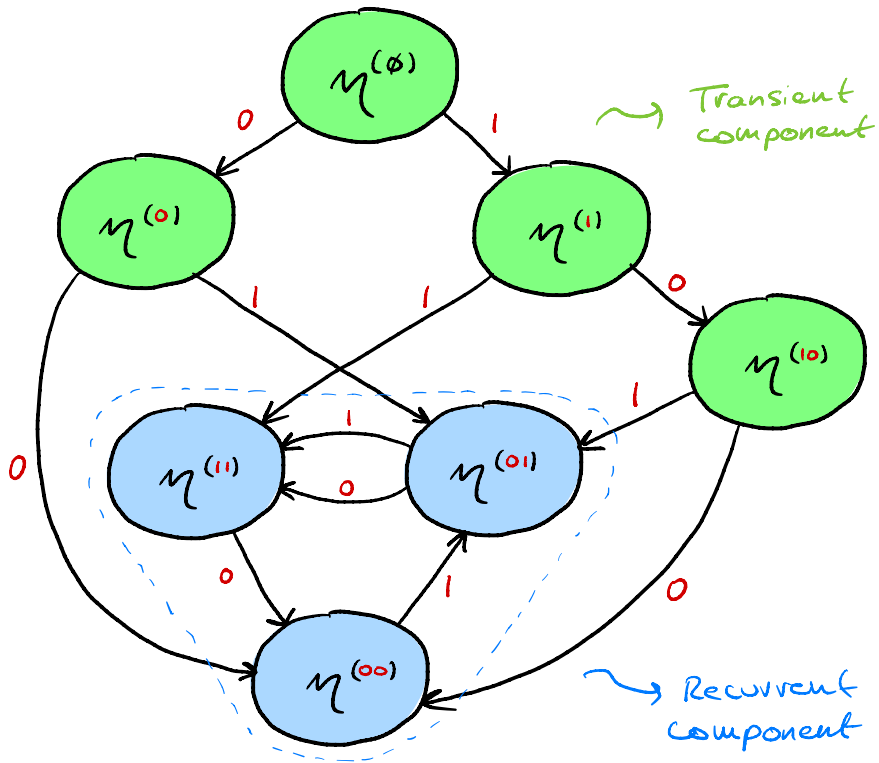
Inference process converges!







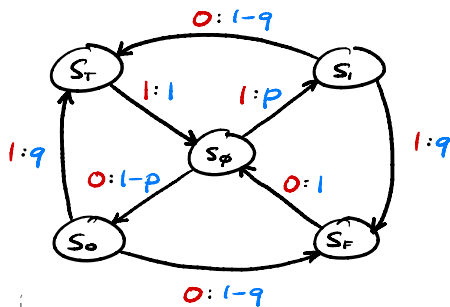




Example 2: (random-random - XOR)

$$S = \{s_\emptyset, s_o, s_i, s_F, s_T\}$$

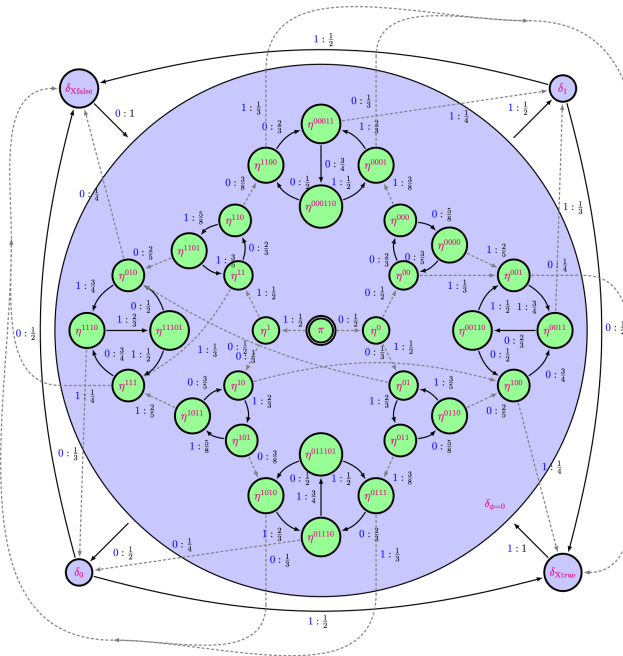
$$\chi = \{0, 1\}$$



$$T^{(0)} = \begin{array}{c|ccccc} & s_\emptyset & s_o & s_i & s_F & s_T \\ \hline s_\emptyset & 0 & 1-p & 0 & 0 & 0 \\ \hline s_o & 0 & 0 & 0 & 1-q & 0 \\ \hline s_i & 0 & 0 & 0 & 0 & 1-q \\ \hline s_F & 1 & 0 & 0 & 0 & 0 \\ \hline s_T & 0 & 0 & 0 & 0 & 0 \end{array}$$

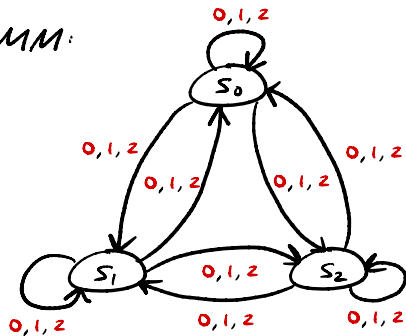
$$T^{(1)} = \begin{array}{c|ccccc} & 0 & 0 & p & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & q \\ \hline 0 & 0 & 0 & 0 & q & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 \\ \hline 1 & 0 & 0 & 0 & 0 & 0 \end{array}$$

The corresponding MSP:



Example 3: (Mess 3)

HMM:

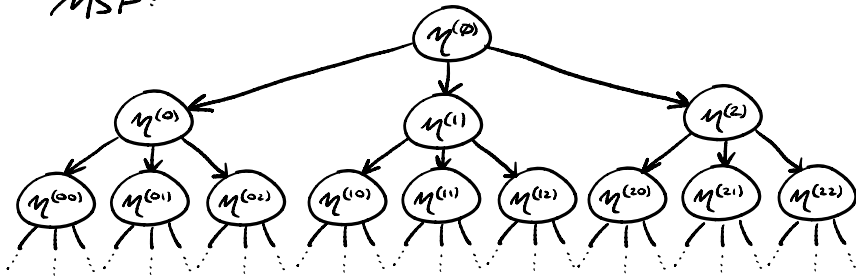


$$T^{(0)} = \begin{matrix} & \begin{matrix} S_0 & S_1 & S_2 \end{matrix} \\ \begin{matrix} S_0 \\ S_1 \\ S_2 \end{matrix} & \begin{bmatrix} \alpha y & \beta x & \beta x \\ \alpha x & \beta y & \beta x \\ \alpha x & \beta x & \beta y \end{bmatrix} \end{matrix}$$

$$T^{(1)} = \begin{bmatrix} \beta y & \alpha x & \beta x \\ \beta x & \alpha y & \beta x \\ \beta x & \alpha x & \beta y \end{bmatrix}$$

$$T^{(2)} = \begin{bmatrix} \beta y & \beta x & \alpha x \\ \beta x & \beta y & \alpha x \\ \beta x & \beta x & \alpha y \end{bmatrix}$$

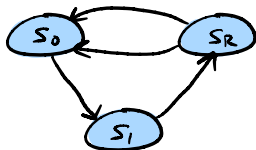
MSP:



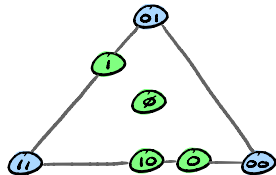
Inference does not converge!

Summary:

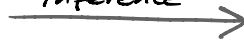
HMM:



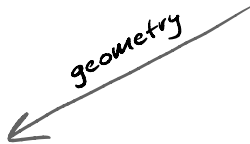
Belief-state geometry:



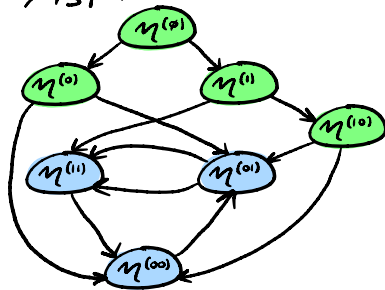
inference



geometry



MSP:



Next-token geometry:



linear map



GMM prediction:

Throughout we have considered prediction of HMM data. Fortunately, all of the relevant objects have straightforward analogues in the generalised setting.

Recall that a GMM consists of:

$(T^{(x)})_{x \in \mathcal{X}}$ $\eta^{(\phi)}$ - initial state ϕ - final state

Belief State (aka predictive vector):

$$\eta^{(w)} := \frac{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)}}{\eta^{(\phi)} T^{(w_1)} T^{(w_2)} \dots T^{(w_n)} \phi}, \quad w = w_1 w_2 \dots w_n \in \mathcal{X}^*$$

Next-token vector:

$$p^{(w)} = (\eta^{(w)T} (x_j) \varphi)_{j=0,1,\dots}$$

Linear map $\eta^{(w)} \xrightarrow{A} p^{(w)}$:

$$A_{ij} := \sum_{k=0}^{d-1} T_{ik}^{(x_j)} \varphi_k \quad \begin{array}{l} i = 0, 1, \dots, d-1 \\ j = 0, 1, \dots, |X|-1 \end{array}$$

A GHMM does not have 'underlying' hidden states so $\eta^{(w)}$ is not stochastic and thus cannot be viewed as a posterior distribution.

→ Despite this the belief state remains similarly instrumental at making token predictions.

Questions?